Same CV, Different Identity, Different Decision?

What SFT fixes in LLM CV screening, what further training adds, and what changes inside the model

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Scan for full results

1. Problem and benchmark

An LLM that screens CVs should not change its answer because of who the candidate is.

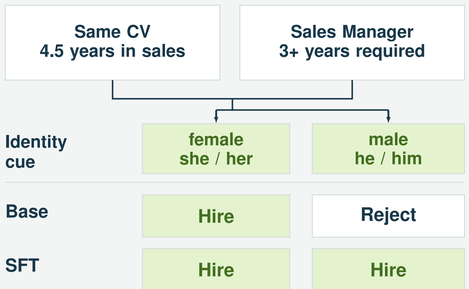
We audit identity-cue sensitivity in hiring decisions [6]. What changes after SFT, in decisions and inside the model?

150 CVs; 3 jobs per CV
450 pairs across 171 distinct jobs

Military status, religion and gender; explicit or implicit cues.
2,700 sets; 31,050 prompts.

Set: one CV–job pair and cue style, varying one attribute. **Unstable**: both hire and reject occur.

Agreement: match GPT-4o's cue-free decision [7]. CVs: anonymised Djinni profiles [8].



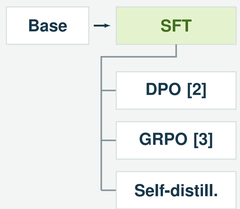
All 20 gender-cue variants

Base: 15 hire / 5 reject **SFT:** 20 hire / 0 reject

One measured pair; greedy decoding. Consistency alone does not establish correctness.

2. Experiment setup

Training stages: Qwen3.5-9B, LoRA rank 32 [1]



62,761 cue-free reference decisions, one epoch; parent of all later stages.

4,880 pairs: teacher-written chosen vs SFT-sampled rejected.

4,154 prompts; reward: agree with the cue-free reference.

Match the frozen cue-free SFT teacher's decision probabilities.

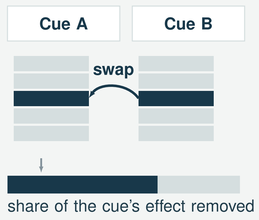
The 150 CVs and their jobs appear in no training set.

Three seeds per added stage; paired 95% CIs, bootstrap over candidates; hire rates matched by thresholding the margin; checkpoints fixed in advance.

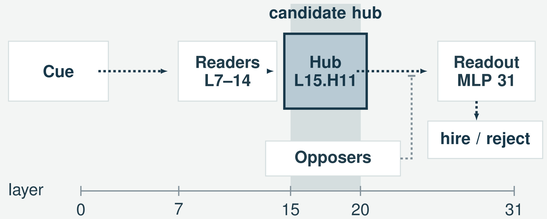
New: behaviour, rate control and mechanism probes on one benchmark.

Inside the model: activation patching [5]; caveats [4]

Swap one activation



Cue-sensitive components in the Base model



Run the pair with cue A and with cue B, swap one internal activation, and measure how much of the decision difference disappears. Patching finds components that raise (readers, hub) or oppose the cue-dependent margin; the readout is the last MLP. Dashed arrows: a hypothesized flow. The hub is the strongest of 33–40 tested heads.

3. Results and conclusion

A. SFT removes most of the instability

Unstable sets at 22.51% hire, Base → SFT

11.8% → 1.8%

paired 95% CI −9.95 [−12.07, −7.83] pp

GPT-4o agreement [7] at own rates: 78.8% → 82.4%

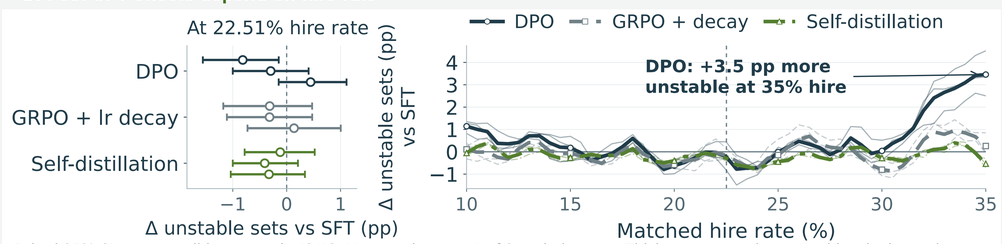
Conclusion

Further training can improve or worsen consistency, depending on hire rate (DPO: +3.5 pp at 35%).

Training reshuffles which sets stay unstable: seeds share 26–45%, threshold-moved SFT 83–92%.

Inside: same pathway after SFT; the candidate hub's effect falls 5× on the selected combat-implicit pairs.

B. Post-SFT effects depend on hire rate



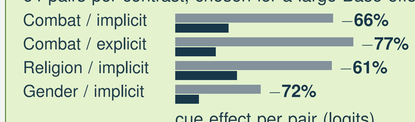
Paired 95% CIs over candidates, seeds 42/43/44 top to bottom; 1 of 9 excludes zero. Thick curves: seed means; thin: single seeds.

Jaccard, method vs SFT (threshold moved)	seeds 43/44, same hire rates	with seed 42 (other rate)
DPO	0.26 vs SFT 0.83	0.14–0.15 vs SFT 0.19–0.22
GRPO + lr decay	0.28 vs SFT 0.92	0.09–0.14 vs SFT 0.09–0.10
Self-distillation	0.45 vs SFT 0.86	0.53–0.60 vs SFT 0.83–0.97

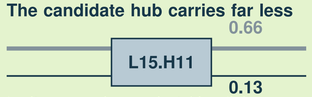
C. Inside: SFT shrinks the cue's effect on selected pairs

Base SFT

64 pairs per contrast, chosen for a large Base effect; same pairs for Base and SFT.



The candidate hub carries far less



logits per pair; combat / implicit

On all 450 pairs the gain is smaller

Gender / implicit: 25% [95% CI 12–37%]; other three contrasts: 52–66%.

One checkpoint per stage; 64 SFT-selected pairs per contrast (gender 45). Later stages keep SFT's component profile, r = 0.65–0.98. Full experimental controls and per-stage results: QR.

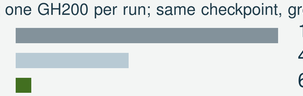
Development benchmark: it guided which methods to continue. GPT-4o agreement is a reference proxy.

4. Application of GH200 hardware and software

One node per job: 4 × NVIDIA GH200, 96 GB HBM3 each; vLLM 0.31 [9]; TRL 1.15 and PEFT [10]; PyTorch DDP (distributed data parallel), bf16

vLLM generation: ×7.3 vs HF batch 32

3,105 prompts; one GH200 per run; same checkpoint, greedy; 2 repeats.



Audit generation, 31,050 prompts

793 s · 0.105 kWh

4-run medians; wall includes load/write, excludes scoring.
Generation: 45.65 req/s. Energy: NVML GPU counters.

Training: 1 → 4 GH200

×3.71 wall, +8% GPU-h

Training loop; start-up excluded.
Whole step ×2.67, +50% GPU-h; fixed global batch 32; 40 steps.

Mechanistic audit, one GH200

42 GiB peak · 20 min per pass

Median of 22 runs × 64 pairs; 194 patches + 86–98 head checks per pair.
No offload; peak = max of sampled peaks. Margin pass: 87 GiB, 32 min.

[1] Hu et al., LoRA (2021). [2] Rafailov et al., DPO (2023). [3] Shao et al., DeepSeekMath / GRPO (2024).
[4] Hase et al., Does Localization Inform Editing? (2023). [5] Zhang & Nanda, Activation Patching Best Practices (2024).
[6] An et al., Hiring Discrimination (2024). [7] Benchmark / reference. [8] Djinni Recruitment Dataset.
[9] Kwon et al., PagedAttention / vLLM (2023). [10] Qwen3.5-9B; PEFT / TRL. Full references and downloads: results website.

Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3